

FOREST FIRE

PREDICTION & SIMULATION

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A Hybrid Deep Learning and Dynamic Pathfinding Architecture for
Uttarakhand



April 20, 2026

Forest Fire Spread
Prediction &
Simulation



THE ABSTRACT.

Forest fires are a major threat to nature, wildlife, and human lives. The Forest-Fire Prediction & Simulation Platform is a smart, web-based tool designed to help authorities manage, predict, and prevent these disasters effectively. At its core, the project uses Artificial Intelligence (Deep Learning) to predict exactly how and where a wildfire will spread. It looks at real-world environmental factors like temperature, humidity, wind speed, and rainfall, combined with satellite images, to calculate the fire's future path.



**Interactive
Fire Simulation**

The system creates a visual simulation of how the fire will grow over the next few days, allowing officials to see potential danger zones ahead of time.



**Safe
Evacuation
Routing**

The platform automatically calculates the safest and fastest paths for rescue teams or civilians, actively avoiding areas where the fire is predicted to spread.



**Instant
Analysis**

Government bodies or disaster management agencies can upload raw satellite data straight into the system to get immediate predictions and rescue routes.

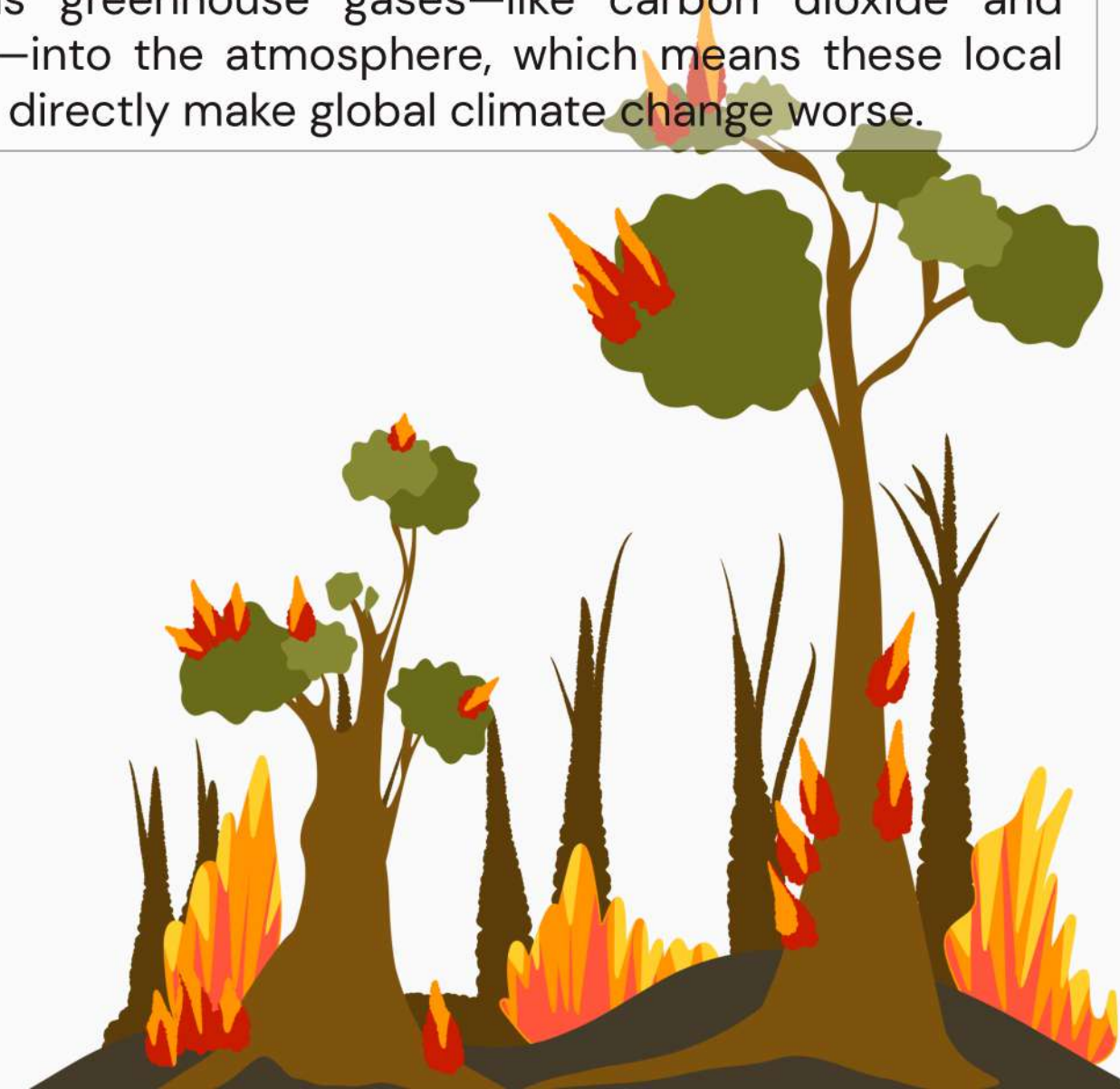
INTRODUCTION.

Forest fires are one of the most dangerous natural disasters in India, especially in sensitive mountainous regions like Uttarakhand, Himachal Pradesh, Madhya Pradesh, and the Western Ghats. When a forest fire breaks out, it causes massive and long-lasting damage to nature. Wild animals lose their homes, and endangered species are put at great risk. Furthermore, without the roots of trees to hold the earth together, the soil easily washes away. This loss of nutrients completely disrupts the delicate natural balance of the local environment.

In recent years, forest fires have become much more frequent and intense. This is largely caused by a changing climate, which has brought hotter temperatures, longer dry spells without rain, and human activities shifting closer to untouched forest lands. Unfortunately, our current methods for monitoring these fires, such as sending forest rangers on foot patrols or relying on basic daily weather reports, are simply not fast or accurate enough. These older methods cannot predict exactly where a new fire might spark, nor can they accurately track how fast the flames will spread across uneven mountain terrain.

To effectively fight back against these fast-moving disasters, there is an urgent need for a smarter, data-driven solution. By combining real-time weather information, satellite maps, and Artificial Intelligence, creating a modern forecasting system becomes possible. This technology can analyze a landscape to calculate exactly where a fire is most likely to start and simulate how it will spread under changing wind and weather conditions. Equipping forest departments with a proactive, high-tech tool like this provides them with the early warnings they need to organize quick emergency responses, map out safe escape routes for civilians, and ultimately stop the flames before they cause further irreversible damage.

Beyond the destruction of nature, these fires cause immense financial and social problems. Valuable resources like timber, medicinal plants, and other forest products go up in flames, causing serious financial damage to the government and to the local villagers whose livelihoods depend on the forest. The social impact is equally devastating, as spreading fires frequently force villagers to flee their homes entirely. Even for those further away, the thick smoke creates terrible air quality that leads to serious health issues like asthma and bronchitis. On a much larger scale, forest fires release massive amounts of dangerous greenhouse gases—like carbon dioxide and methane—into the atmosphere, which means these local disasters directly make global climate change worse.



AIM'S & OBJECTIVE.

The primary goal of this project is to build a highly accurate, smart computer system that can pinpoint where forest fires are likely to spread, realistically simulate how they will spread over time, and automatically draw safe emergency escape routes for people in danger.

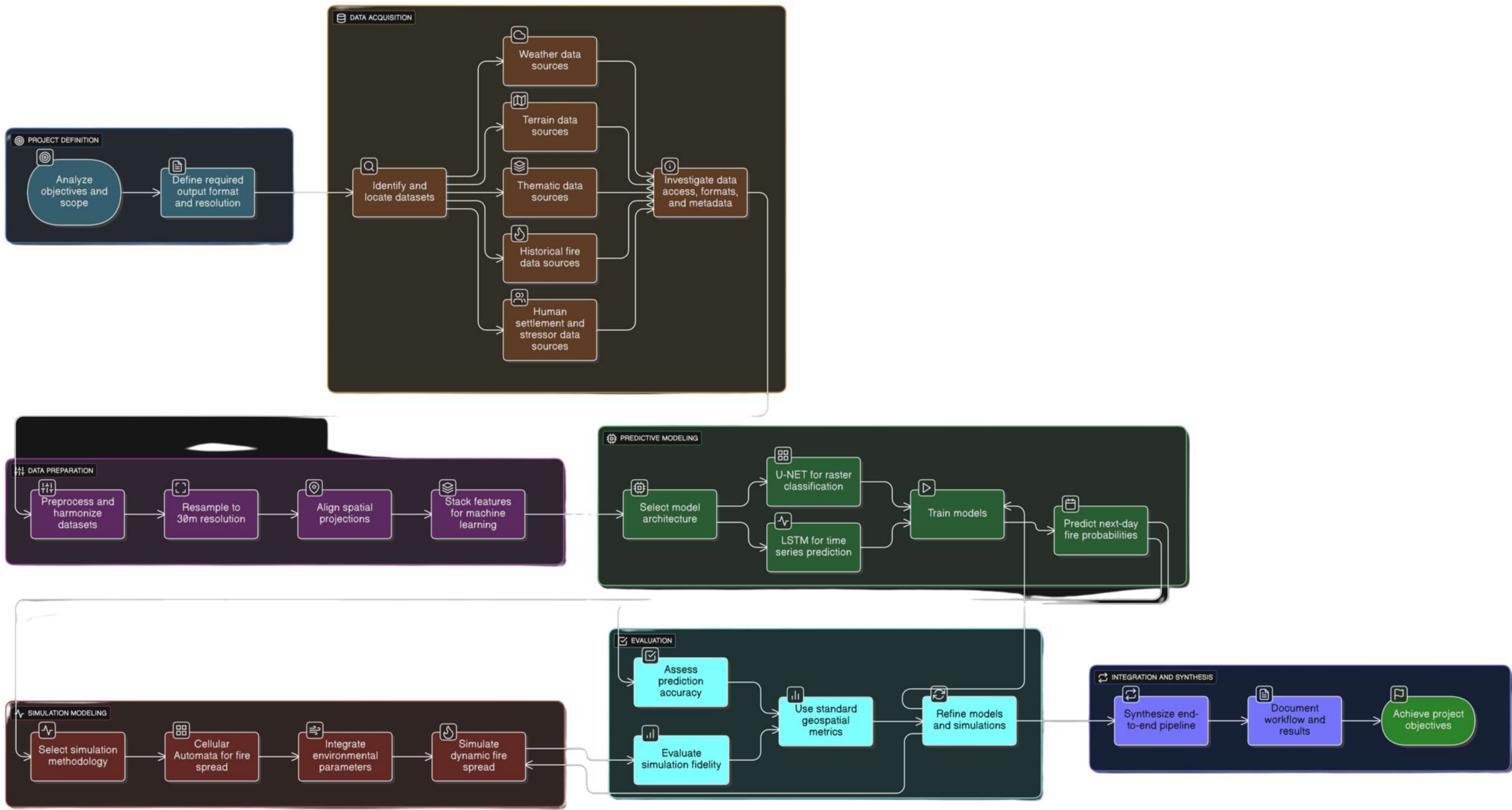


- 1. Beat the "Identity Trap:** Stop traditional AI from simply guessing that the fire will stand perfect
- 2. Build a Smart Fuel Map:** Engineer an AI that focuses strictly on finding the most flammable "fuel" in the forest by studying weather patterns, soil dryness, and mountain slopes.
- 3. Simulate Real-World Physics:** Develop a fast-running logic engine that mimics the physical rules of nature—like ensuring fire spreads in realistic "cigar shapes" downwind and travels rapidly up steep hills.
- 4. Design Dynamic Dual-Routing:** Create a smart GPS system that calculates safe street routes, and if roads fail, calculate safe off-road fallback paths to ensure people can escape.

LITERATURE REVIEW.

Reference Paper Title	Key Themes	Drawback	Findings/Conclusions	Implications for Travel Industry
Wildland and Forest Fire Prediction in Thailand using Satellite Data — Phankrawee	Satellite hotspots, meteorology, ML	Study limited to Thailand data	Effective hotspot detection with ML pipeline	Enables near-real-time alerts for tours/flights
Forest Fire Spread Simulation and Fire Extinguishing Visualization Research	3D visualization, CA model, suppression tactics	Focused on visualization, not large-scale forecasting	Useful for training and testing tactics	Useful for evacuation drills and site safety planning
Ubiquitous GIS-Based Forest Fire Susceptibility Mapping Using AI Methods	GIS + remote sensing, mobile maps	Depends on local spatial data quality	High predictive performance; mobile maps work well	Provides risk maps to avoid high-risk routes
Simulation of forest fire spread based on artificial intelligence	ANN model, weather, terrain	Trained for limited area / samples	Good spread prediction (high accuracy)	Helps issue warnings and plan safe routes

PROPOSED SYSTEM.



TECHNICAL APPROACH.

High-Performance Tools for Real-Time Calculations



FRONTEND

Next.js 15

React

Tailwind CSS

Leaflet Maps



BACKEND

Python

FastAPI

RESTful APIs

High-Speed Compute



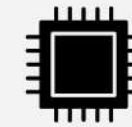
MACHINE LEARNING

PyTorch

U-Net Architecture

Satellite Imagery

Risk Prediction



ALGORITHMS

Cellular Automata

D* Lite Pathfinding

OpenRouteService

Physics Simulation

SYSTEM ARCHITECTURE.

01



02



03



04

DATA COLLECTION

Gathering real-time climate data (temperature, wind) and high-resolution mountain topography maps.

AI RISK MAPPING

Our custom U-Net AI model analyzes land data to create a map highlighting flammable "danger zones."

PHYSICS SIMULATION

Applying mathematical engines to simulate wind effects and fire rushing uphill over time.

USER DASHBOARD

Visualizing movement on an interactive website with automated safe escape route calculation.

FEATURES IMPLEMENTED.

01.

Interactive Fluid Simulation

You can click anywhere on the web map to start a mock fire, drag sliders to change wind speed, and instantly watch the fire stretch out downwind across the mountains

02.

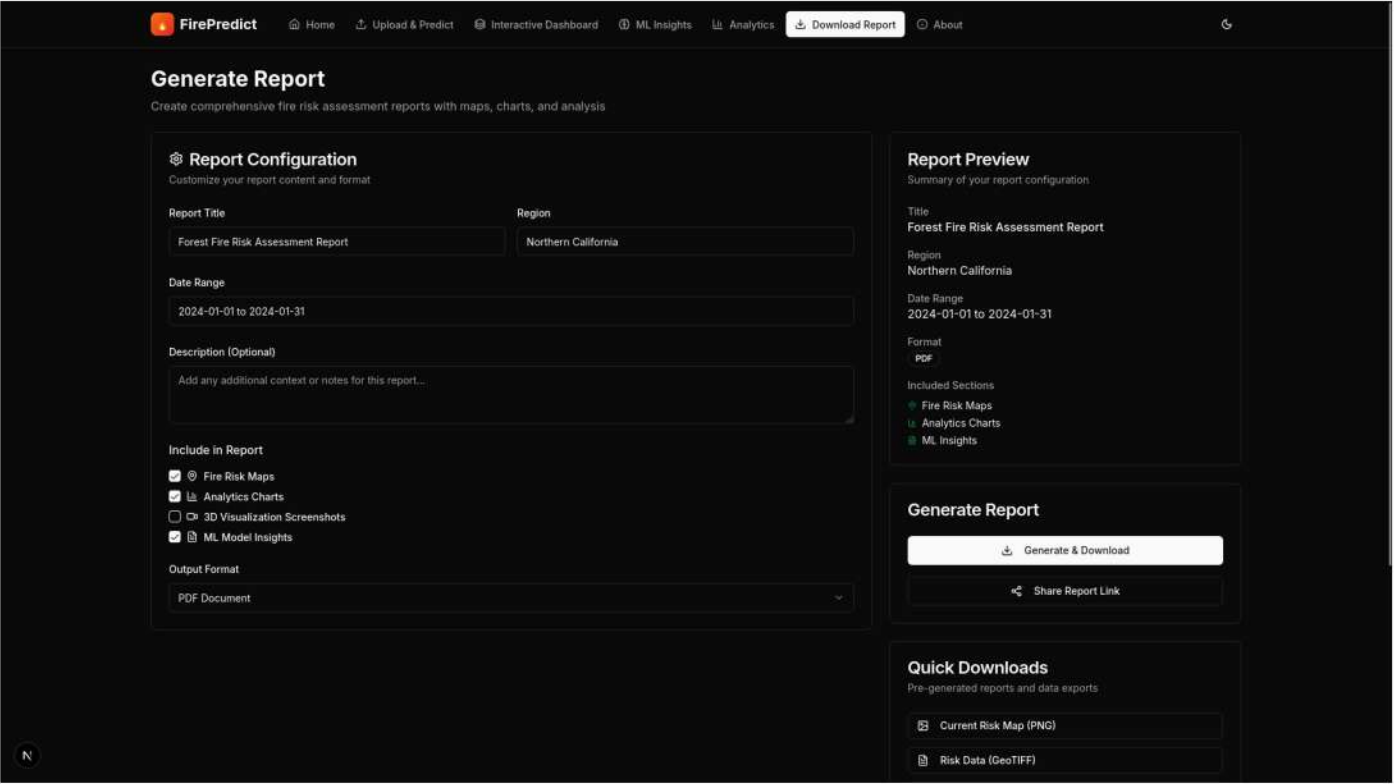
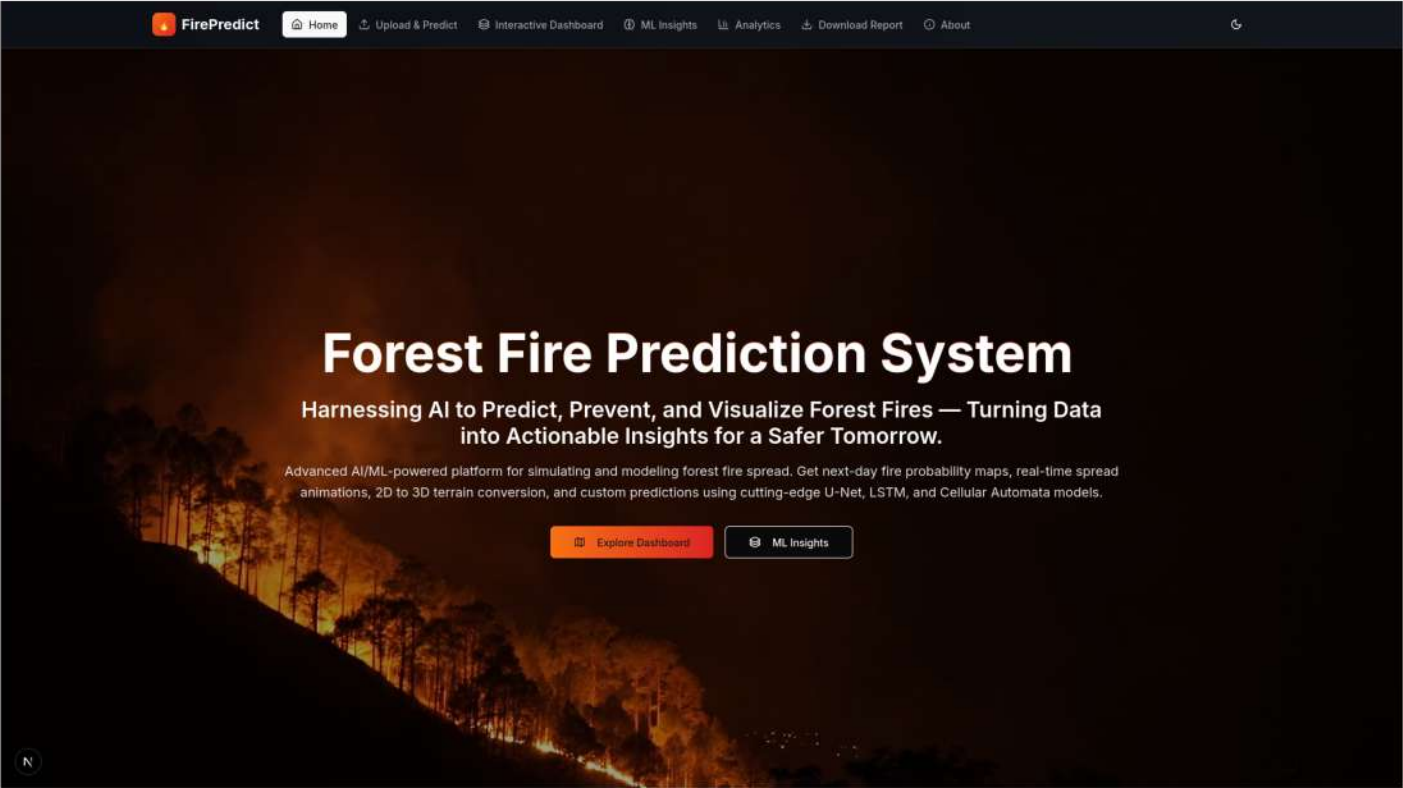
Smart Road Snapping (OpenRouteService)

If a user drops a rescue pin deep in the forest where there are no roads, the system executes an "infinite search" algorithm to automatically snap their start point to the nearest valid highway.

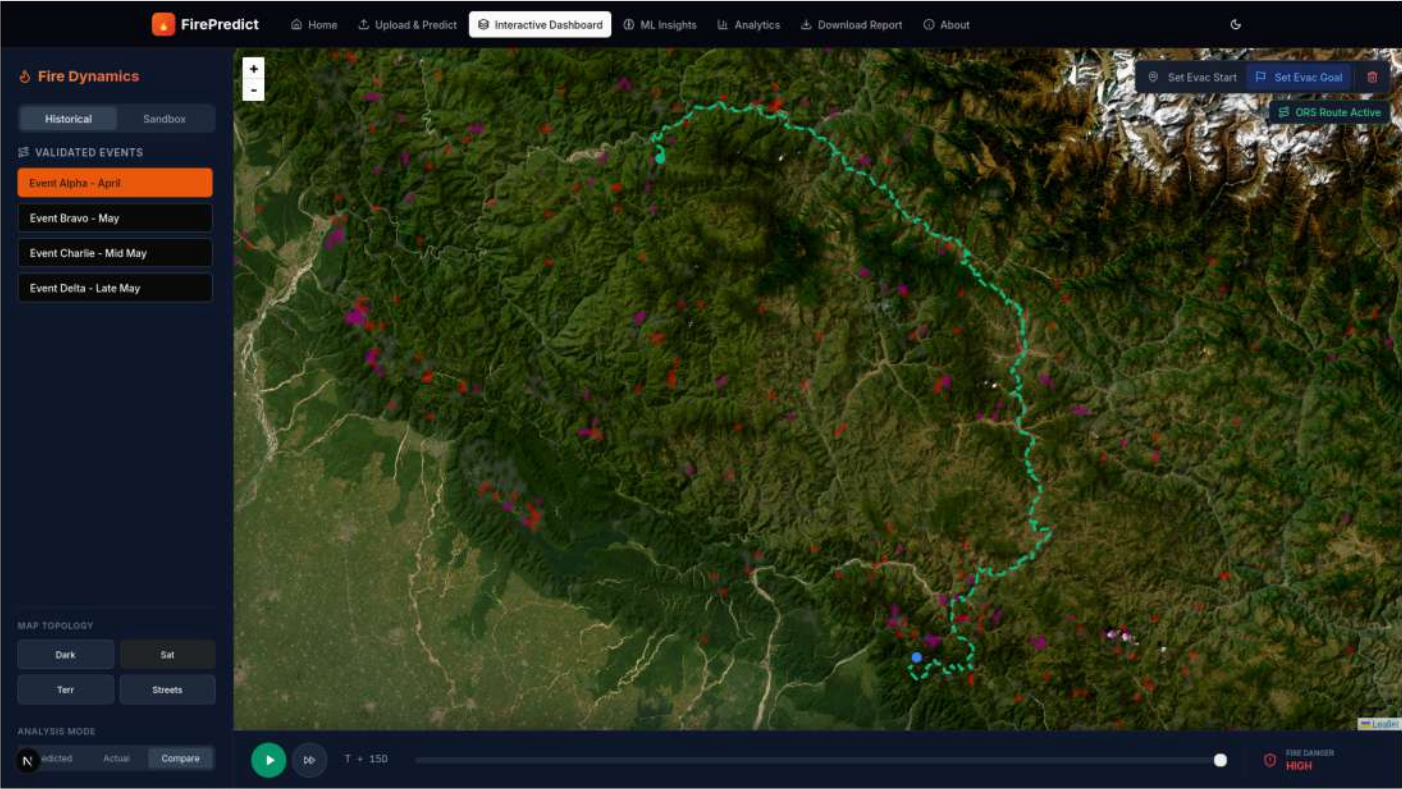
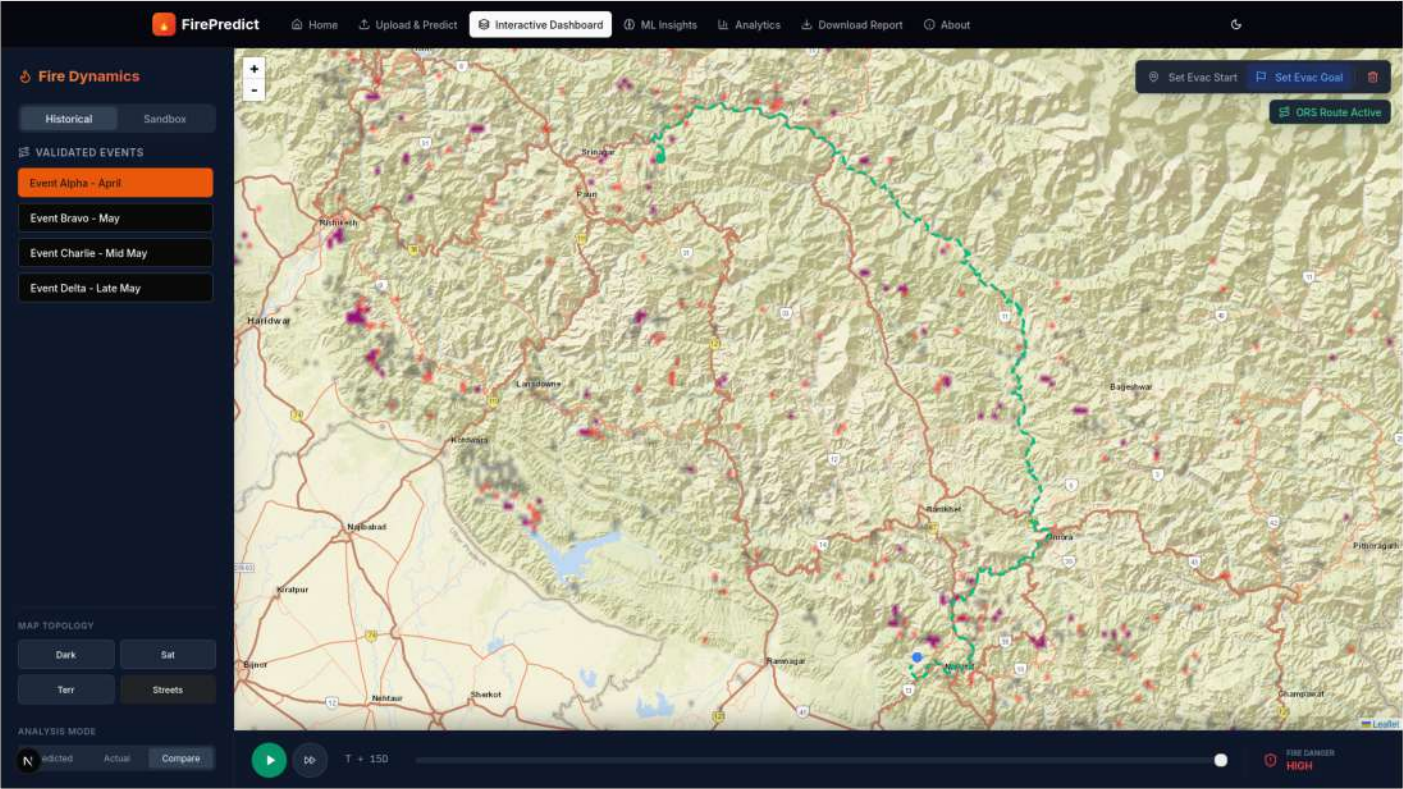
03.

Off-Road Fallback (D* Lite)

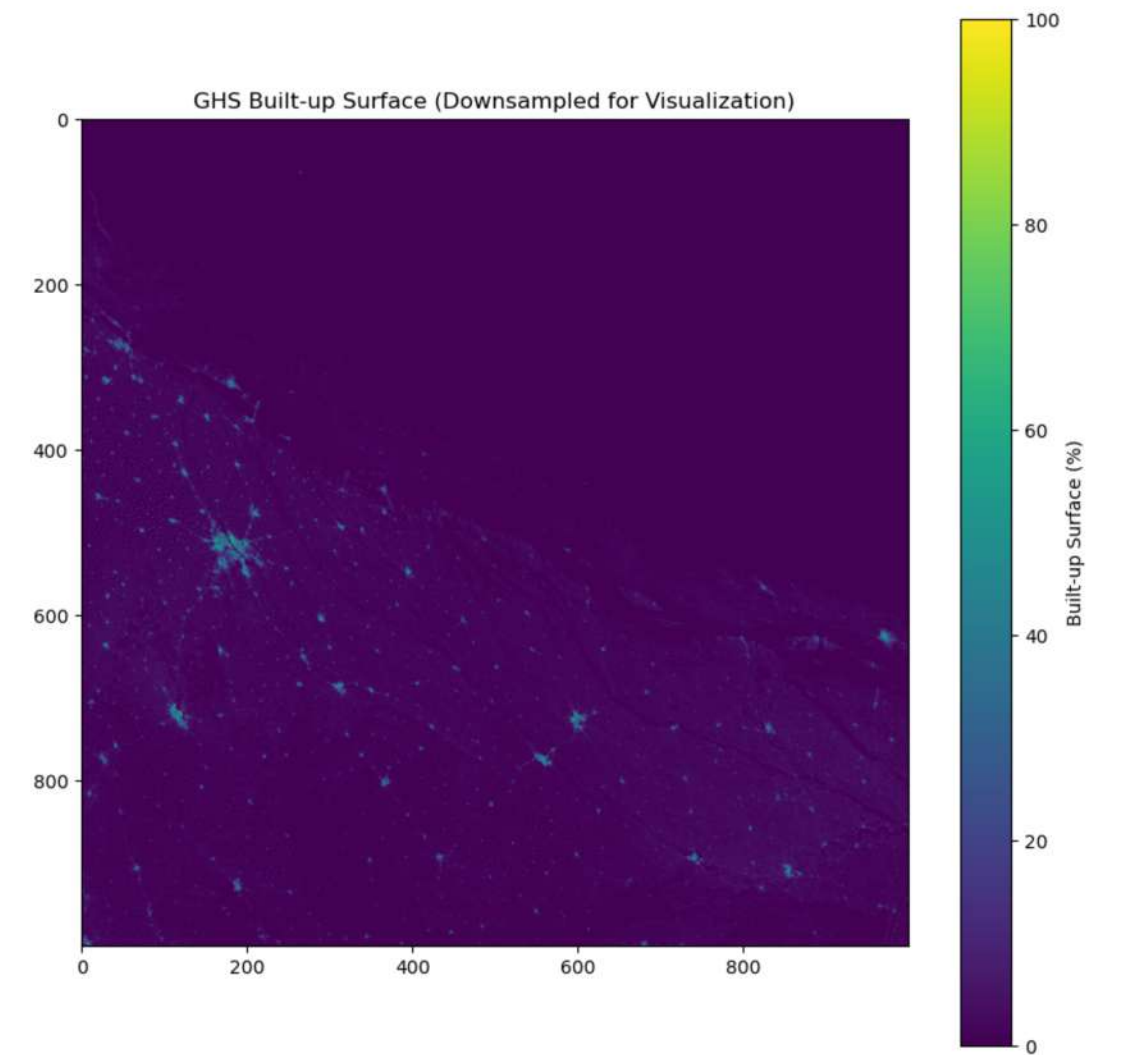
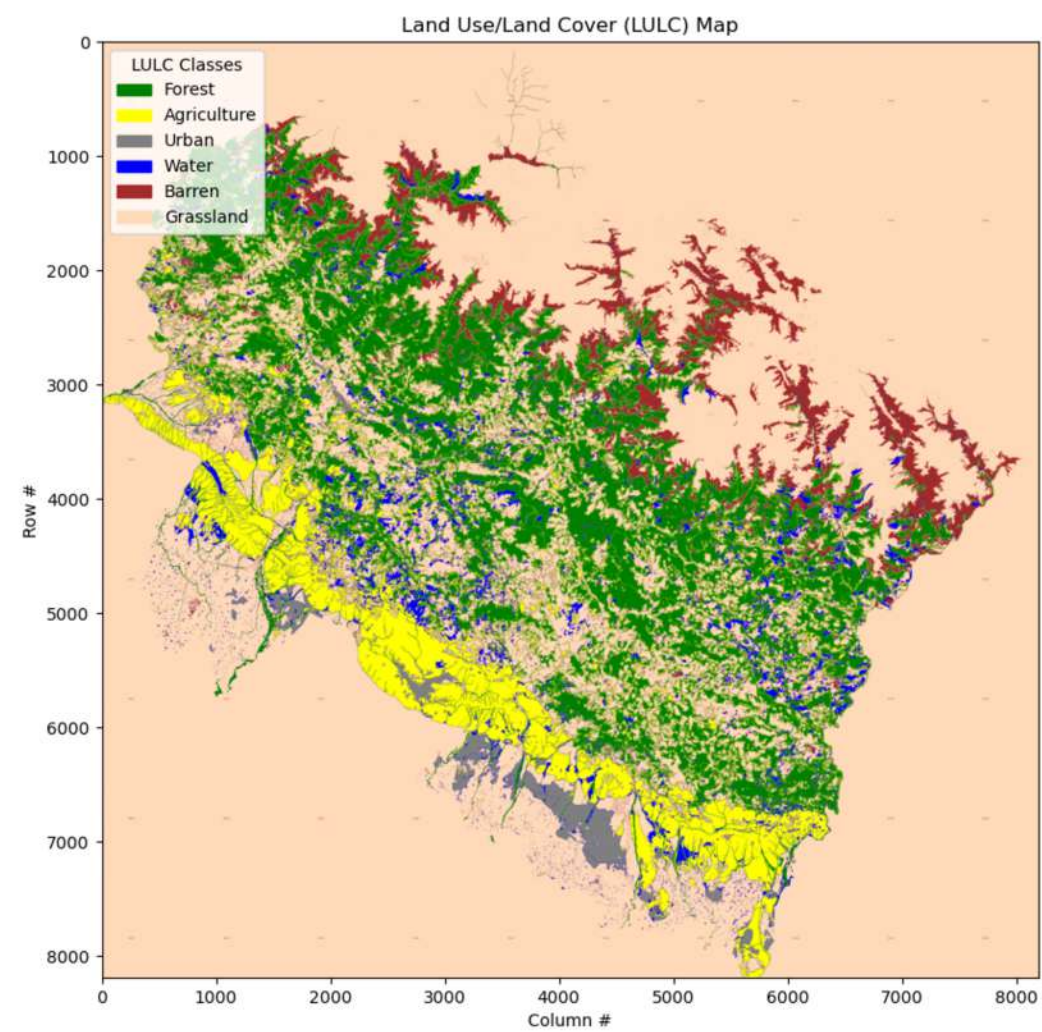
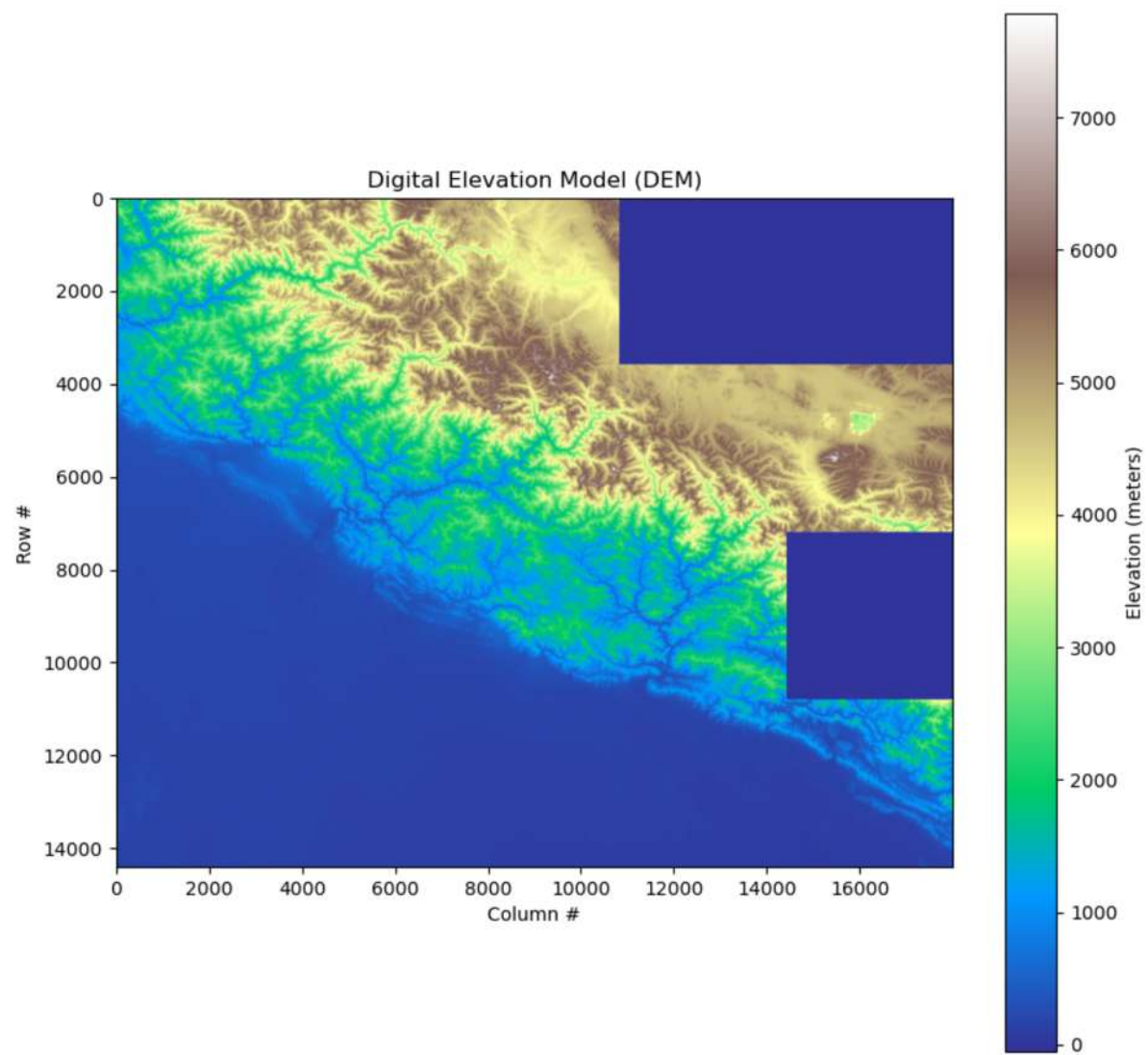
If real-world road networks fail, our Python backend runs an advanced grid-search. It searches backward from the destination, easily recalculating detours through the forest itself to dodge the simulated flames.



OUR
PAGES



OUR RESULTS



ERA5-Land

Attr	FullForm
e	Evaporation
tp	Total Precipitation
---	---
tcc	Total Cloud Cover
stl1	Soil Temperature Level 1
slt	Soil Type
swvl1	Volumetric Soil Water Level 1
cvh	High Vegetation Cover
cvl	Low Vegetation Cover
tvh	Type of High Vegetation
tvl	Type of Low Vegetation
u10	10m U Component of Wind
v10	10m V Componenet of Wind
d2m	2m Dewpoint Temperature
t2m	2m Temperature

[illegible]

MODIS (Historical Fire Data)

Attr	Description
scan, track	These values relate to the dimensions of the fire pixel. They can be used to calculate the actual size of the pixel on the ground.
acq_date, acq_time	The date and time when the satellite detected the fire. This is essential for temporal analysis and linking fire events to specific weather data.
satellite, instrument	The satellite (Terra or Aqua) and instrument (MODIS) that captured the data.
confidence	A value from 0-100% indicating the quality of the detection. High-confidence detections are more reliable.
bright_t31	The brightness temperature of the fire pixel in band 31 (11.0µm), in Kelvin. This is used for fire detection and characterization.
frp	Fire Radiative Power, measured in megawatts. This is a measure of the fire's energy and intensity, which can be used to estimate biomass burning.
daynight	Indicates whether the detection was made during the day or night.
type	Categorizes the type of fire, such as "1" for a vegetation fire.

DATASET

Final-MODIS

	latitude	longitude	brightness	scan	track	acq_date	acq_time	satellite	confidence	bright_t31	frp	daynight	type
0	29.3724	77.9882	315.9	1.1	1	2016-04-01	812	Aqua	41	294	7.1	D	0
1	29.7752	78.459	315.7	1.1	1.1	2016-04-01	812	Aqua	43	286.6	8.3	D	0
2	29.6941	78.7024	320	1.1	1.1	2016-04-01	812	Aqua	33	298.7	10.8	D	0
3	29.7108	78.9152	319.6	1.2	1.1	2016-04-01	812	Aqua	61	303.6	8.5	D	0
4	29.4864	78.5851	320.6	1.1	1.1	2016-04-01	812	Aqua	58	302.1	7.2	D	0
5	29.847	79.3609	321.7	1.2	1.1	2016-04-01	812	Aqua	72	302.7	13.4	D	0
6	29.6781	78.7292	327.6	1.1	1.1	2016-04-01	812	Aqua	79	299	21.3	D	0
7	29.6767	78.718	319.5	1.1	1.1	2016-04-01	812	Aqua	33	298.1	10.1	D	0
8	29.6962	78.5195	313.5	1.1	1.1	2016-04-01	812	Aqua	24	290.5	5.4	D	0
9	29.7094	78.9038	322	1.2	1.1	2016-04-01	812	Aqua	69	305.9	11.4	D	0

Source: (<https://firms.modaps.eosdis.nasa.gov/country/>)

YOU GOT THIS

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Forest Fire Spread Prediction &
Simulation

Rajiv Gandhi Institute of Technology,
Mumbai

This platform bridges the gap between AI prediction and
actionable disaster response to protect lives and
ecosystems.

THANK YOU

